THE SPACELAB PROGRAM AND RELATED LEGAL ISSUES +

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The purpose of this paper is to illustrate the impact of the Spacelab on law and security in outer space. To this effect Spacelab will be considered from the technical, political and legal points of view.

I. What is Spacelab from the technical point of view?

The Spacelab is a modular space laboratory carried out in the cargo bay of the Space Shuttle Orbiter, offering a great degree of flexibility to accommodate various experiments. Spacelab is therefore an integral part of the NASA Space Transportation System (STS). It is carried by the orbiter, to which it remains attached throughout the flight and is fully reusable.

Spacelab can be assembled in a variety of configurations. The basic elements of the assembled unit include pressurized module sections and unpressurized pallets. The module provides the working environment for the payload specialists who will man the laboratory in orbit, while the pallet is a platform on which instruments are mounted. Also included in the program is the delivery of an Instrument Pointing System.

The pressurized module is a cylindrical unit made of aluminum alloy. It can have one or two segments, according to mission requirements. Each segment is 4 meters in diameter and 2.7 meters long. When the two segments are used together the module can carry a payload of up to 4.6 tons and provides a usable working volume of 22 cubic meters of experiments. One of the two segments is known as the core module because it houses the essential sub-systems—monitoring equipment, electrical supplies, computers and thermal regulation and some experiments. The other segment is the experiment module housing experiments only.

Unlike the experiment module, the pallets are not pressurized and are, indeed, directly exposed to the space vacuum. A Pallet Train can consist of up to five segments and can carry approximately 9000 kilograms of instruments. The payload could include telescopes and antennae or radars. These may either function automatically, or may be controlled from the module, the orbiter or the ground. The Spacelab module and a pallet can be used together for a particular mission, or, either one of them may be used separately. Hence the unit has considerable flexibility as a research medium.

The Spacelab is linked to the orbiter by a tunnel, 1 meter in diameter, which enables the payload specialists to enter the Spacelab and return to the Shuttle and share the astronauts' living accommodation. Normally two Payload specialists will man the Spacelab.

What are the results of the Spacelab program achieved so far? To date the Spacelab program as seen from the European side is nearing its completion and has reached some interesting results for Europe - to be evaluated against the 950 millions of dollars spent by the European partners. The following events evidence some of these results:

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+ The views expressed in this article are those of the author and not necessarily those of the European Space Agency.

The first Spacelab element arrived at Cape Kennedy in mid-December 1980.

Delivery of the Spacelab flight Unit was made in two batches - the first was delivered to NASA in December 1981 and solemnly handed over to the United States in the presence of Mr. George Bush, Vice-President of the United States, the 2nd of February 1982. The second batch was delivered in mid 1982.

The European payload for the first flight was integrated at ERNO, Bremen, under the management of the ESA facility called SPICE in Germany. The payload was flown to the United States during the summer of '82 for final integration with the United States payload at KSG.

The first flight of Spacelab in the Orbiter (SL-1) is due to take place in September-October of 1983 and the second Spacelab mission (SL-2) in November 1984. In addition one dedicated flight of Spacelab has been booked by Germany (D1 mission) for May 1985.

It is understood that the flights of Spacelab after SL-1 will be under the sole responsibility of NASA and any European participation will be on a commercial basis.

The training of the European selected as payload specialist for SL-1 is in good progress. In addition, one ESA astronaut is completing training as a mission specialist at KSC and could therefore fly on further Spacelab flights if NASA so decides.

A lot of specialized equipment has been developed in Europe as part of the program. In particular it should be mentioned that during the second and third orbital flight tests of the Orbiter a pallet manufactured in Europe carried a set of scientific instruments provided by NASA.

Finally it is worth noting that NASA purchased, in accordance with its obligation, a second Spacelab and additional equipment required for the Spacelab program. In January 1980 NASA signed a contract with ESA for the purchase of the second Spacelab; the contract was extended in May to include a second Spacelab Instrument Pointing System.

II. What is the Spacelab program from the political point of view?

Europe participates in the major United States space program which is the United States Space Transportation System. In this cooperative venture Europe undertook in 1973 to finance, design and develop a manned space laboratory. NASA would operate the laboratory as part of the United States Space Transportation System. It is the biggest cooperative venture between ESA and NASA. For Europe it is of particular significance since it heralds the entry of the European scientific and technological communities into the realms of manned space flight.

Europe is entirely responsible for the funding, the design and development of the Spacelab. NASA is to provide the launcher and the Space Shuttle which will carry the laboratory in orbit for each mission and bring it back to Earth once a mission is terminated. On the European side, ESA is fulfilling its obligations in the development of this major program, which will open new fields for basic and applied scientific research in space. Nine Member States (Belgium, Denmark, France, Germany, Italy, Netherlands, Spain, Switzerland and the United Kingdom) and one State with associate member status (Austria) are pooling the technical, financial and intellectual resources in the interests of the program.

Germany is the main contributor to this program, providing over 50% of the budget. The prime contractor is the German firm VFW-ERNO in Bremen. Some 40 companies in Europe have been involved in construction of the Spacelab. At the height of the development phase an industrial workforce of about 2000 was employed on the program.

The Spacelab program includes the right for Europe to obtain half of the first mission free of charge and to fly a European payload specialist in the first flight. During this flight half of the payload will be provided by the United States and half by Europe. It also obliged NASA to purchase from ESA a second set of Spacelab hardware identical to the first one provided to NASA by ESA free of charge.

III. What are the legal issues arising from the Spacelab program?

The Spacelab program, being a cooperative program, should be seen on two separate counts. Firstly, Spacelab is being constructed on a cooperative basis by a number of European States grouped within the European Space Agency (ESA)[1]; and, secondly, it is being incorporated in the Space Shuttle, which itself constitutes the first element of the space transportation system developed by the United States. It will be seen, then, that Spacelab involves two sets of legal relations - one set bearing on its development and the other on its utilization. These relations encompass not only the Europeans and the Americans but also third parties, in particular the users. To the intrinsic complexity of these relations must be added the fact that since Spacelab is not designed to fly on its own, it is closely dependent on the Shuttle, a situation that entails major legal consequences. Two such legal consequences are as follows: A. Spacelab as an international cooperative program. This will provide the occasion for describing the legal scaffolding that has been erected to permit its development and utilization. B. Spacelab as an integral part of the Space Shuttle. This will lead one to identify the legal differences, vis-a-vis space law, that distinguish Spacelab from the Space Shuttle.

A. Spacelab as an International Cooperative Program

The development of Spacelab by ESA for delivery to NASA - which is responsible for its integration in the Shuttle - and the use of Spacelab on Shuttle flights are the subject of a series of legal texts that define the relations between the various parties concerned. Spacelab has given rise to two series of texts, one consisting of agreements between Europe and the United States, and the other of intra-European agreements.

The first category comprises:

> The Agreement between the Government of the United States of America and certain Governments, Members of the European Space Research Organization (now the European Space Agency), for a Cooperative Program concerning Development, Procurement and Use of a Space Laboratory in conjunction with the Space Shuttle System. This Agreement, which was signed on 14 August 1973, lays down the general principles of a transatlantic cooperative program for the development of Spacelab and defines the program's objectives and general characteristics.[2]

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[1]*Convention for the Establishment of a European Space Agency*, BASIC TEXTS OF THE EUROPEAN SPACE AGENCY, Vol. I at A-6, entered into force, Dec. 9, 1976.

[2]*Agreement Between the Government of the United States of America and Certain Governments, Members of the European Space Research Organization, for a Cooperative Programme Concerning Development, Procurement and Use of a Space Laboratory in Conjunction with the Space Shuttle System*, BASIC TEXTS OF THE EUROPEAN SPACE AGENCY, Vol. II, at G9b, entered into force, Aug. 14, 1973.

> In conformity with the foregoing Agreement, a Memorandum of Understanding between NASA and ESRO (now ESA), which was also signed on 14 August 1973, sets out the modalities governing the implementation of the program and specifies, inter alia, the respective responsibilities of the two Agencies.[3]

The second category comprises the intra-European agreements that preceded, or have followed, these two texts. They comprise:

> An arrangement between certain Member States of ESRO (now ESA) and the Organization, itself, which was signed on 15 February 1973. The purpose of this arrangement was to determine the rules under which the European countries were prepared to discharge the commitments that they would be entering into with the United States and NASA. In the context of the present paper it should be noted that this arrangement includes a provision that gives the Agency ownership of Spacelab.[4]
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> A series of texts relating to the constitution of the European structure for assuring integration of Spacelab in the Space Shuttle. The structure in question is an Agency team (SPICE-Spacelab Payload Integration and Coordination in Europe) set up in the Federal Republic of Germany.[5]

Spacelab utilization, like Spacelab development, has given rise to two series of agreements. In the category comprising agreements between the American and European partners one again finds the two texts already mentioned:

> The inter-governmental Agreement lays down the general principle that the United States Government will make the Space Shuttle available to the Europeans for their Spacelab missions (experiments and applications) on either a cooperative or a cost-reimbursable basis. Special rules are provided for the first Spacelab unit. It will pass under full American control once it is delivered and its first flight will be a cooperative mission in which Europeans will participate as Spacelab payload specialist.
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> The Memorandum of Understanding between the two Agencies lays down, inter alia, the conditions under which subsequent Spacelabs will be acquired by the United States. The Memorandum establishes the principle that the United States must refrain from separate and independent development of any Spacelab that would constitute a duplication of the one developed by the Agency.

In the category comprising intra-European agreements, the Member States of the Agency have agreed to execute a Spacelab utilization program within the framework of the Agency.[6] Implementing Rules were adopted by the participants and approved by the ESA Council at a later stage on 10 February 1982. For the execution of this program, which, for the first time, is confined to the first flight of the first Spacelab, a number of texts have also been adopted with regard to:

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[3] *Memorandum of Understanding Between the National Aeronautics and Space Administration and the European Space Research Organization for a Cooperative Programme Concerning Development, Procurement and Use of a Space Laboratory in Conjunction with the Space Shuttle System,* BASIC TEXTS OF THE EUROPEAN SPACE AGENCY, Vol. II at G9c, entered into force, Aug. 14, 1973.

[4] *Arrangement Between Certain Member States of the European Space Research Organization and the European Space Research Organization Concerning the Execution of the Spacelab Programme,* BASIC TEXTS OF THE EUROPEAN SPACE AGENCY, Vol. II at G9a, entered into force, Aug. 10, 1973.

[5] *See generally,* BASIC TEXTS OF THE EUROPEAN SPACE AGENCY, Vol. II at G9a—e.

[6] Council Resolution of October 4, 1977; Declaration by the Participants on December 12, 1977.

The provision by certain Member States of general-purpose or special instruments for the conduct of the experiments; and,

The conduct of the experiments by the Agency on behalf of the experimenters.

By the date of the first Spacelab mission it will be necessary to settle certain outstanding issues. In the first instance, those that stem from the legal status of Spacelab in relation to the Space Shuttle must be dealt with.

B. Spacelab as an Integral Part of the Space Shuttle

It emerges very clearly from the texts just mentioned that it is impossible in practice to dissociate the Spacelab from the Space Shuttle. This is true despite the fact that these two elements of the same space transportation system have different owners. Because the notion of ownership of a space object leads, under space law, to a number of consequences, it is necessary that a solution be found to the situation created by the discordance between the Arrangement and the Agreement. Either the Agency will seek to exercise its right of ownership over the first Spacelab - even after it has been delivered to NASA - and in that case the appropriate conclusions will have to be drawn at all levels; or the Agency will refrain from doing so, in which case formal transfer to NASA of the title to ownership will have to be envisaged. This issue is currently being examined by the European Space Agency and will be the subject of discussions with the U.S. authorities with a view to clarifying the situation.

1. Ownership of Spacelab

As already indicated, the Arrangement between European States stipulates that the elements of the first Spacelab developed under the program are the property of the Agency, acting on behalf of the participants, as are the facilities and equipment acquired for the execution of the program. Article VII sec. D of the Agreement between the United States and Europe, on the other hand, stipulates that this first Spacelab shall be delivered to the American Government and that it shall be placed under the full control of the latter which may, in addition, decide to modify it. Although the first mission is a cooperative one, the American Government alone will have the responsibility for its successful execution, both from the technical viewpoint and from the aspect of liability for any damage occurring to the Spacelab itself (with the exception of damage resulting from the launch, flight or descent of the Space Shuttle). Lastly, the American Government will have unrestricted use of the first Spacelab for other missions, free of cost. In other words, the Agency, which is the owner of the first Spacelab cannot exercise its ownership rights once it has delivered the unit to NASA.

2. Registration

According to the Convention on Registration of Objects Launched into Outer Space: "When a space object is launched into orbit or beyond, the launching State shall register the space object by means of an entry in an appropriate registry. . . ."[7] This

[7]Convention on Registration of Objects Launched into Outer Space (hereinafter "Registration Convention"), January 14, 1975, 28 U.S.T. 695, T.I.A.S. No. 8480.

leads one to ask whether Spacelab is a "space object." In the absence of any definition of the term in the various space agreements, we think that the answer is "Yes", notwithstanding the fact that Spacelab is dependent on Space Shuttle during its flight. Thus the Agency, having launched a space object, would be entitled to register it independently from the registration of the Space Shuttle as a whole. This latter registration is naturally the responsibility of the United States.

3. Jurisdiction Over Personnel and Materials

This issue covers civil, criminal and disciplinary jurisdiction as well as intellectual property rights. According to the 1967 Space Treaty, registration of a space object entitles the registering State to retain jurisdiction and control over the object and over any personnel thereof.[8] But this provision only applies to States, and the Agency could not, therefore, avail itself of it. Moreover, operational imperatives would render inoperative any rights that might be recognized with respect to Spacelab, since once it is integrated in the Space Shuttle - and all the more so when it is in flight - it is an integral part of the latter.

4. Liability for Damage

This issue can arise under various aspects depending on whether one is dealing on the one hand with damage caused to the United States, the Agency or its Member States, or on the other hand with damage caused to third parties. In the case of the first group, such damage would be covered by the existing agreements between the American and European partners. This leaves the question of damage caused to third parties. The latter might invoke the Conventions on liability[9] and on registration[10] as a means of directly involving the liability of the Agency, as owner of Spacelab.

5. U.S. Domestic Legislation

One should also mention that since 1973, some U.S. domestic legislation governing persons and property involved in STS flights has been adopted by the U.S. Government for its own purposes. The following provisions will have to be taken into consideration when trying to solve the issues referred to above:

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[8]Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, Jan. 27, 1967 *[1967]*, 18 U.S.T. 2410, T.I.A.S. 6347, 610 U.N.T.S. 205 (effective Oct. 10, 1967).

[9]Convention on International Liability for Damage Caused by Space Objects, March 29, 1972, 24 U.S.T. 2389, T.I.A.S. No. 7762.

[10]Registration Convention, *supra* note 7.

A regulation on objects likely to be carried on the STS flights.[11]

A regulation on the authority of the STS Commander.[12] Under this regulation, the authority of the Commander applies to all persons aboard the Shuttle, including Spacelab, whether they are of American nationality or not.

A regulation on the safety of personnel. The purpose of this regulation was to establish criteria and procedures to ensure that personnel having access to the STS should meet safety standards.[13]

A regulation on insurance in the event of damage to third parties. The NASA Act was amended by the Congress in order to introduce a new section 308 authorizing NASA to provide, in certain cases, liability insurance for the users. The customer can obtain an insurance covering his liability and that of the American Government for damage to third parties up to a ceiling of 500 million dollars. Beyond this amount NASA remains liable. As regards relations between users and NASA and between users, the risk theory prevails; each one is liable for his own damage and waives recourse against other parties.

Finally it is worth mentioning that NASA and ESA have already discussed guidelines for the conduct of Spacelab experiments using humans as test subjects in order to achieve common standards.

Conclusion

Besides this legal aspect, the implementation of the Spacelab program may be a basis for a certain amount of reflections on both sides on how difficulties arising from the execution of a cooperative program may be solved. Europe has certainly encountered a lot of problems from the fact that, for obvious reasons, the Spacelab program was totally dependent on the development of the orbiter which was NASA's responsibility. Thus, the European partners had to comply not only with NASA's technical demands concerning the Spacelab itself - which were a consequence of NASA's control of the Space shuttle program as a whole - but also with the consequences in terms of the calendar of NASA's decision on the Orbiter development program. In both cases the effect was a significant increase (or 40%) of the overall cost of the Spacelab development program. On the other hand, the European partners found that the only reward they had, that is the right to fly half of the first Spacelab payload free of charge and to sell only one other Spacelab - with some additional equipment - was really too small in comparison with their financial effort. In addition, no further extension of this cooperative venture has been agreed to yet, which is contrary to the expectations of the European States.

The future of Spacelab is an important issue in the planning of the European Space Agency program in the years to come. ESA has already decided to start a Spacelab follow-on program comprising a free flying European Retrievable Carrier (Eureca). Also included are some improvements to the Spacelab itself provided that NASA provides increased power to the payload and enables longer duration Shuttle missions (Now 7 to

[11] 43 Fed. Reg. 204 (October 26, 1978).

[12] 45 Fed. Reg. 47, at 14, 845 (March 7, 1980).

[13] 44 Fed. Reg. 131 (July 6, 1979).

12 days). In addition, ESA would also like to participate with NASA in the studies of large space stations.

If the decision is taken to continue the US/Europe cooperation in such a direction, one may be sure that numerous legal problems will arise, on the one hand for improving the mechanism of such cooperation, and, on the other hand for solving the legal issues which we are already facing. I am confident that after Apollo/Soyuz, Intercosmos and Spacelab flights, international manned space flights will be more and more frequent in the years to come. This will certainly be a challenge for all people involved: politicians, scientists, technicians, administrators and, of course, lawyers.